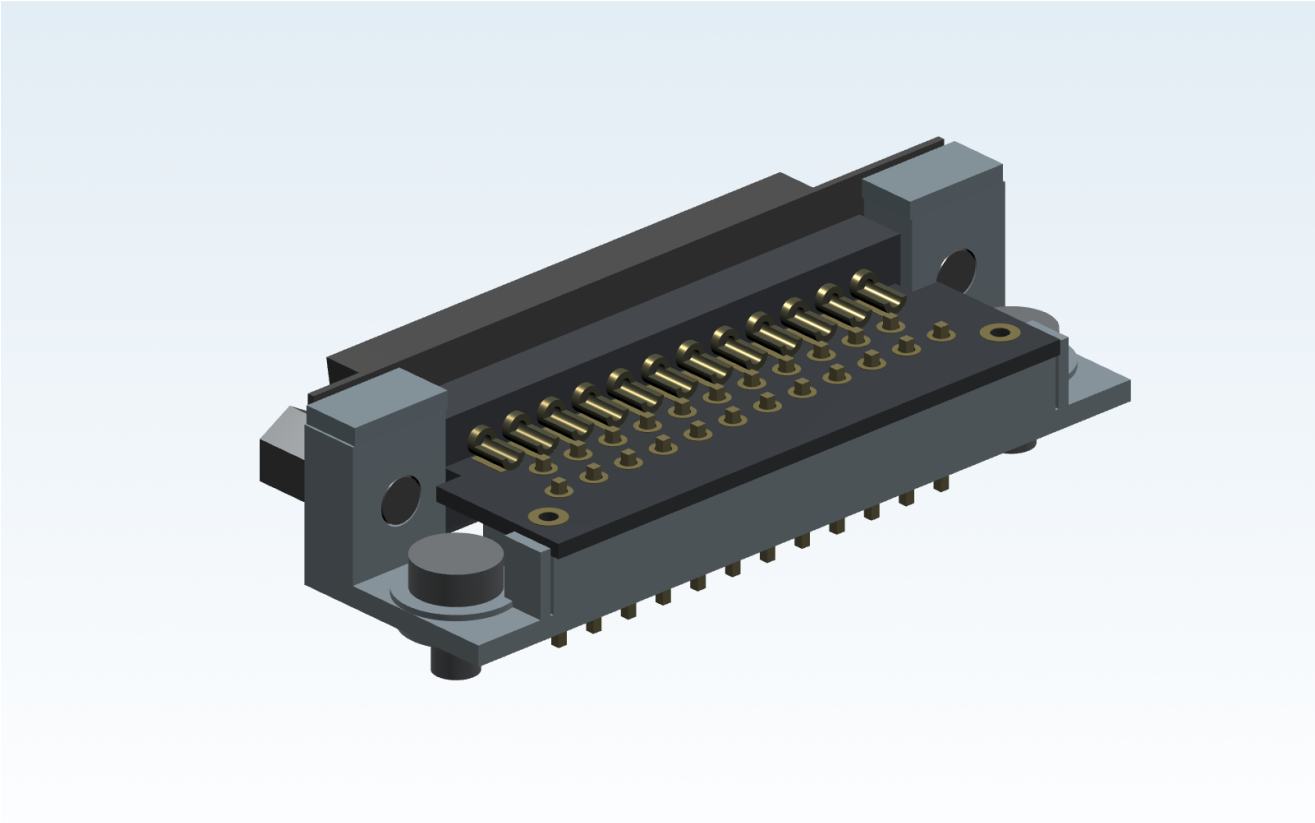
DB23 replacement assembly

Genuine DB23 solder-cup connector + two-layer PCB + printed pin-support holder

DB23 replacement - assembled CAD preview

UK Computing Limited | Revision 0.3 | Nominal dimensions in millimetres



Black: adapter PCB. Grey: printed holder. Gold: contacts and supported motherboard pins.
FIT PROTOTYPE v0.3. Solder-cup connector and bought hardware are unmeasured references.

A common printed holder supports 23 individual motherboard pins and the connector flange. Separate male RGB and female floppy PCBs retain the original Amiga cable interface. The 3D views use an illustrative, unmeasured solder-cup connector and fastener geometry.

PCB outline	**40.00 x 12.00 mm, notched front**
PCB thickness / underside height	**1.00 mm / 5.75 mm assumed**
Holder width / rear position	**52.00 mm / Y15.60 mm**
Motherboard pin grid	**2.77 pitch; 2.84 row spacing**
Shell anchor centres	**44.27 mm; Y9.50 mm**

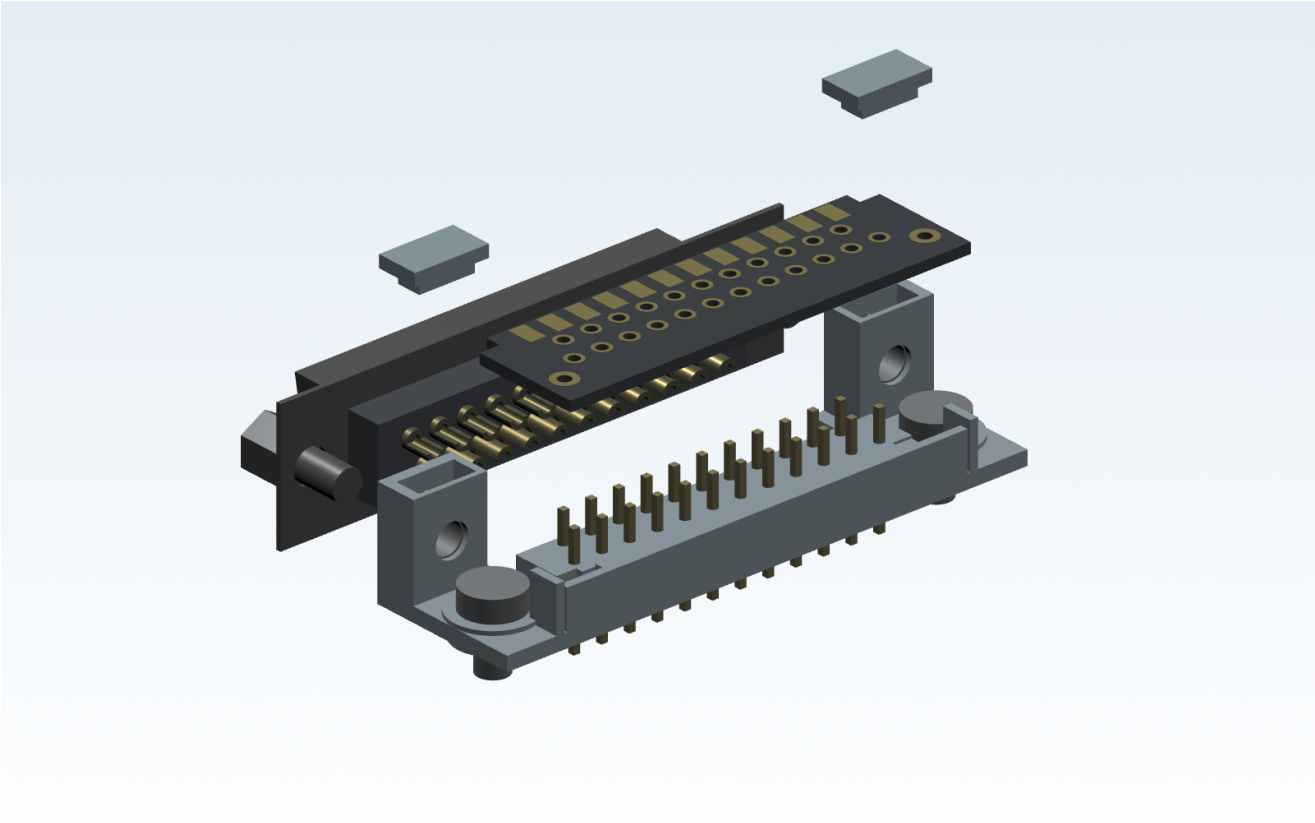
Prototype status: both boards pass native KiCad PCB DRC and exact connection checks. The holder is a closed printable solid. These checks do not establish fit, strength or soldering heat performance with the purchased parts. Complete the measurement worksheet before PCB manufacture.

Exploded view and assembly

PCB and nut retainers lifted; connector moved forwards for visibility

DB23 replacement - exploded CAD preview

UK Computing Limited | Revision 0.3 | Nominal dimensions in millimetres



PCB and nut keepers lifted; connector moved forwards. Pins remain in the printed guide block.
Exploded offsets are illustrative. Shell bonding wires are omitted; they remain required.

Parts per connector assembly

Solder-cup connector	**1 DB23 male OR female**
PCB and straight pins	**1 matching PCB +23 pins**
Printed parts	**1 holder; 2 shims; 2 optional caps**
Cable jackposts and captive nuts	**2 each; M3 rear / 4-40 front**
Motherboard anchors and shell bond	**2 anchor sets; separate shield bond**

Build order

1. Measure the connector and pins; print the hole-fit coupon.
2. Print and trial-fit the holder, nuts and mandatory 0.60 mm anchor shims.
3. Set the 23 pin lengths and solder the PCB/DB23 using a heat-safe support.
4. Check the complete unpowered assembly, shell bond and all 23 connections.

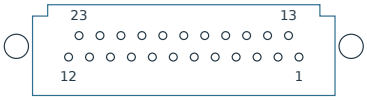
Fit references and measurements

Full-size pin templates below: print at 100% / Actual size, with no page scaling

Motherboard top view; flange faces the top of the page

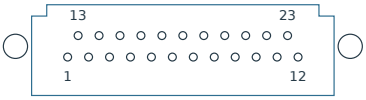
Pin 1 is mirrored between variants. PCB corner notches clear the printed nut towers.

MALE / RGB



Blue outline: actual adapter PCB

FEMALE / FLOPPY



Blue outline: actual adapter PCB

50.00 mm scale check

Pin-hole test coupon

The supplied STL includes trial bores from 0.75 to 1.10 mm and the complete 23-hole pattern. A 0.64 mm square pin has a 0.905 mm diagonal: 0.80 mm is not a finished clearance bore. Start with the coupon and actual pins; determine the required drilled or reamed fit.

Measure before freezing the PCB and holder

Purchased DB23 part number / gender	_____
Gap between cups / usable cup length	_____
Flange to rear body / cup-end depth	_____
Actual pin section / finished guide bore	_____
Jackpost rear thread / stem projection	_____
Nut AF / thickness / pocket fit	_____
Target motherboard / case clearance	_____

Datum: Y0 is the flange front plane; assumed flange metal spans Y0-0.50. The holder begins Y0.65. Its bottom is Z0.60 above the motherboard; the two 0.60 mm anchor shims preserve the intended height. Jackpost stems and anchor bolt heads must clear one another. See mechanical/MECHANICAL_NOTES.txt for the nominal envelopes.

Sources: original connector grid and reference geometry: iComp DB23 replacement drawings. All solder-cup, nut, printer and fastener assumptions are recorded in source/design_parameters.json. Native electrical and mesh reports are in verification/ and mechanical/.